AUV / ROV SHIP HULL STRUCTURAL INSPECTION REPORT

Autonomous Robotic NDT Survey • Generated: 2026-10-20 13:15:09 • Mission ID: INSP_20261020_122619_7364d0

1. Executive Mission Summary

Metric	Value	Metric	Value
Inspection ID	INSP_20261020_122619_7364d0	Start Timestamp	2026-10-20T12:26:27
Hull Surface Inspected	44.45%	Total Logged Steps	400
Suspected Defects Found	45	Mean Detection Confidence	17.8%
Water Visibility Range	0.50 - 0.50	ROV Battery Remaining	96.8%

2. Suspected Hull Defects Detected

Step	Zone ID	Pose (X, Y) [m]	Defect Category	Confidence	Inspection Status
131	Stern-Keel-Bottom	(46.2, 1.8)	Corrosion	79.8%	Requires Inspection
132	Stern-Keel-Bottom	(45.8, 1.8)	Corrosion	81.1%	Requires Inspection
133	Stern-Keel-Bottom	(45.4, 1.8)	Corrosion	80.9%	Requires Inspection
134	Stern-Keel-Bottom	(45.0, 1.8)	Corrosion	81.6%	Requires Inspection
135	Stern-Keel-Bottom	(44.6, 1.8)	Corrosion	81.6%	Requires Inspection
136	Stern-Keel-Bottom	(44.2, 1.8)	Corrosion	83.0%	Requires Inspection
137	Stern-Keel-Bottom	(43.8, 1.8)	Corrosion	83.9%	Requires Inspection
138	Stern-Keel-Bottom	(43.4, 1.8)	Corrosion	84.2%	Requires Inspection
139	Stern-Keel-Bottom	(43.0, 1.8)	Corrosion	83.9%	Requires Inspection
223	Bow-Keel-Bottom	(9.7, 1.8)	Corrosion	88.2%	Requires Inspection
224	Bow-Keel-Bottom	(9.3, 1.8)	Corrosion	88.4%	Requires Inspection
225	Bow-Keel-Bottom	(8.9, 1.8)	Corrosion	88.4%	Requires Inspection
226	Bow-Keel-Bottom	(8.5, 1.8)	Corrosion	88.3%	Requires Inspection
227	Bow-Keel-Bottom	(8.1, 1.8)	Corrosion	88.2%	Requires Inspection
228	Bow-Keel-Bottom	(7.7, 1.8)	Corrosion	88.6%	Requires Inspection
229	Bow-Keel-Bottom	(7.3, 1.8)	Corrosion	88.8%	Requires Inspection
260	Bow-Keel-Bottom	(6.7, 3.0)	Corrosion	88.7%	Requires Inspection
261	Bow-Keel-Bottom	(7.1, 3.0)	Corrosion	88.7%	Requires Inspection
262	Bow-Keel-Bottom	(7.5, 3.0)	Corrosion	89.0%	Requires Inspection
263	Bow-Keel-Bottom	(7.9, 3.0)	Corrosion	88.6%	Requires Inspection

3. Scientific Limitations & Marine Engineering Disclaimer

SCIENTIFIC & ENGINEERING LIMITATIONS:

1. This automated inspection system identifies visible surface corrosion and cracking patterns using optical computer vision. Internal metallurgical stress, micro-fatigue, and subsurface cracks cannot be measured from RGB camera imagery alone.
2. All model detections are classified as *'suspected defects'* with status *'requires further inspection'*. Confidence scores represent statistical model activation, not calibrated structural failure probabilities.
3. Confirmatory structural integrity assessment requires certified marine surveyor evaluation and complementary non-destructive testing (NDT), such as ultrasonic thickness gauging (UT) or eddy current testing.